

Week 7 Module Project B: Q-Learning with Bayesian Belief Tracking

Acting under partial observability with normalized posterior state

CMPSC 480 - Student Question Deck

Assessment overview

Integrate a from-scratch Bayesian hazard belief with a Q-learning policy.

Coverage: Weeks 4-7: MDPs, Q-learning, approximation, Bayes nets, and inference

Points

20 total

Expected time

10-14 hours

Submission

Repository, evaluation report, and belief traces

Learning objectives

- ▶ Update a categorical belief using Bayes rule.
- ▶ Handle impossible or numerically tiny evidence defensively.
- ▶ Expose belief summaries to a Q-learning state representation.
- ▶ Train and evaluate under several sensor-noise regimes.
- ▶ Ablate belief tracking against a nonbelief baseline.

Requirements checklist

- ▶ Use Python 3.11 and NumPy; do not use a library that implements the assessed algorithm.
- ▶ Keep data, model, optimization, and evaluation responsibilities behind explicit interfaces.
- ▶ Validate shapes, ranges, finite values, empty inputs, and terminal or degenerate cases.
- ▶ Use deterministic seeds and record every configuration used for reported evidence.
- ▶ Include focused tests and a README with exact reproduction commands.

Task 1 - Belief model (5 points)

Implement a categorical prior, sensor likelihood, and normalized posterior over candidate hazard locations.

- a. Write the vectorized posterior update. [2 points]
- b. Hand compute prior $[0.5, 0.5]$ with likelihoods $[0.8, 0.2]$ for one observation. [2 points]
- c. Define behavior when the normalization constant is zero or nonfinite. [1 points]

Task 2 - RL agent (4 points)

Implement terminal-aware Q-learning using only information available to the agent.

- a. Define the observable agent state and prove it excludes the true hidden hazard. [2 points]
- b. Specify epsilon-greedy selection and the terminal Q update. [2 points]

Task 3 - Integration contract (4 points)

Connect observations to posterior updates and posterior state to action selection.

- a. State the within-step order from observation to action to learning. [2 points]
- b. Test that reset restores the declared prior and clears episode-local state. [1 points]
- c. List every component needed to resume training exactly. [1 points]

Task 4 - Noise study and ablation (4 points)

Evaluate policy safety and reward at low, medium, and high sensor noise.

- a. Define safety and performance metrics and run multiple seeds. [2 points]
- b. Ablate the posterior input and interpret the difference. [2 points]

Task 5 - Defensive evidence (3 points)

Test belief and policy behavior on adversarial boundary cases.

- a. Provide four tests and name the defect each exposes. [2 points]
- b. Explain why a high-noise posterior should remain closer to the prior. [1 points]

Edge cases and defensive programming

- ▶ Prior has a negative entry.
- ▶ Evidence has zero likelihood under every hypothesis.
- ▶ Sensor is completely uninformative.
- ▶ Episode begins terminal.
- ▶ True hidden state accidentally enters features.

Submission instructions

- ▶ Submit source, tests, README, configuration, and the requested report.
- ▶ Provide one command that reproduces the primary result from a clean environment.
- ▶ Exclude credentials, generated caches, and unlicensed data.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric - part 1

Category	Points	Required evidence
Belief model	5	BeliefTracker, fixed-sequence oracle tests, and trace table.
RL agent	4	Agent interface, update tests, and reproducible training loop.
Integration contract	4	End-to-end sequence diagram, integration tests, and checkpoint serialization.

Grading rubric - part 2

Category	Points	Required evidence
Noise study and ablation	4	Controlled result table with belief and no-belief baselines.
Defensive evidence	3	Focused tests and a short failure analysis.

Reflection prompt

In 150-250 words:

Describe one case where the belief was mathematically correct but the policy still behaved poorly. Separate inference error from control or learning error.